

Aclara™ Power Line Systems TWACS Speed and Bandwidth





TWACS Has Unique Capabilities for a PLS system that Offer Substantial Improvements in Speed and Performance

- **Group Addressing**
 - TWACS endpoints can be addressed in groups of 64 for meter reads and 256 for outage pinging.
 - A single endpoint can be read in 20 seconds, but a group of 64 can be read in 76 seconds
- **Distribution System Parallelism**
 - TWACS can simultaneously retrieve data from metering endpoints on different substation busses, feeders, and phases, increasing throughput by many multiples.
- **Command Priorities**
 - Need to perform a disconnect or read an individual meter while collecting interval data. No problem. Just set the command priority for the disconnect higher than the interval data read and it execute right away.



The Speed of the TWACS signal

- How long does it take a light bulb to illuminate when you flip on a switch?
 - The TWACS signal is a part of the voltage sine wave so as fast as the sine wave can get to the bulb is as fast as a TWACS command can get to an endpoint!
 - The data returning from the endpoint rides the current sine wave and returns just as fast as the signal gets there!
 - In all, a single “serial” on demand meter read can take as little as 8sec of power line time from start to finish and 20 seconds from initial request to display of data. This is the same speed or faster than RF technologies.
- But what about getting data from thousands of endpoints?
 - TWACS commands are “grouped” together so literally every endpoint on a feeder receives the command and groups of up to 256 endpoints share 6 channels to return their data at the same time!
 - The entire TWACS network is a parallel network where all phases, feeders, busses and substations communicate concurrently!
 - TWACS NG can read 200 meters per minute (or more) per each substation bus. A substation bus with 5,000 endpoints can be read in 25 minutes or less. 100 substations with 500,000 endpoints can still be read in 25 minutes or less.

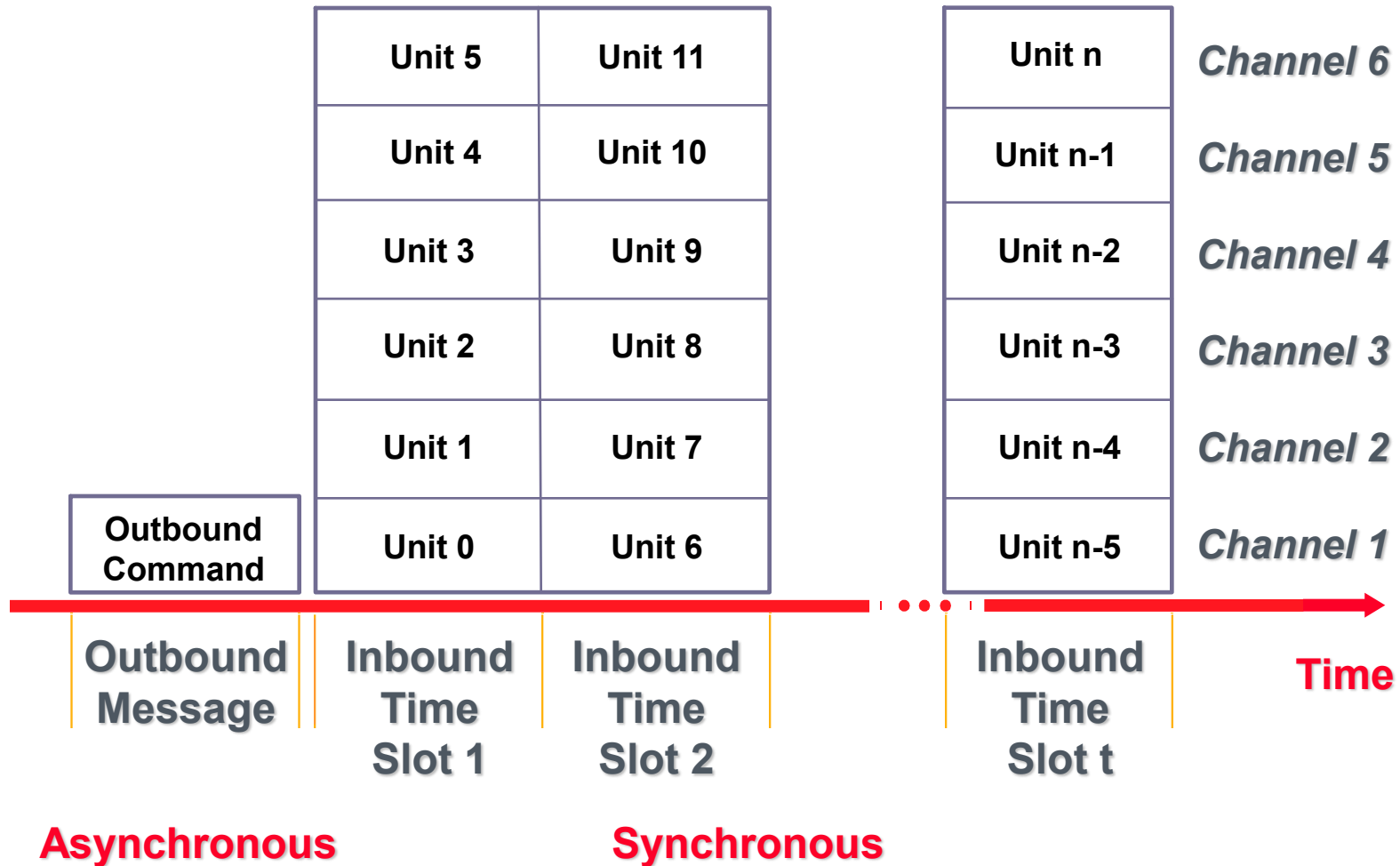


Efficient Use of Groups and Channels

- Group size
 - TWACS protocol supports groups sizes of up to 256
 - Maximize throughput by maximizing group size
 - Master Station typically supports group sizes of 16 and 64 for meter data, and 256 for fast poll pings
- Channels
 - Six TWACS inbound channels
 - Maximize use when using groups sizes of more than six
 - Small ad-hoc groups, e.g., retries – more efficient to use small groups than individually to take advantage of multiple channels
- Group of 64 residential read, 6 channels
 - 63 seconds (60Hz)
 - 76 seconds (50 Hz)



High Performance **TWACS** Retrievals Grouped for Efficiency





Concurrent Use of Substations and Buses

- Substations can operate independently
- Each substation has a command queue of up to 36 commands
- The master station polls each substation periodically to send new commands and retrieve responses, keeping all substations busy
- Independent buses within a substation can operate concurrently

Concurrent Phase

- Within each bus, the system can operate three phases concurrently
- Typical throughput increase of 2.7x using concurrent Phase



Concurrent Feeder

- Within a bus, the system can operate the feeders concurrently
- One outbound addresses multiple feeders
- Address up to 8 or more feeders in parallel
- Potential performance increase of 8x
 - Typically less than 8 feeders on a bus
 - Greatest gain with group meter read commands



Achieving high bandwidth through concurrency

- Outbound
 - Generated per bus/phase – outbound can occur on each bus/phase in the system independently
 - All transponders on the bus/phase hear the transmission
 - The packet is decoded and each transponder decides if the packet is destined for it
- Inbound
 - Each bus is independent in the TWACS system
 - Each feeder is independent
 - Each phase is independent
 - 6 independent channels per phase



Achieving high bandwidth through orthogonal channels, group addressing and timeslot responses

- A single on request read – 1 piece of data
 - The inbound message contains 8 bytes of data
 - Each bit takes 4 cycles
 - $4 \text{ cycles} \times 8 \text{ bits/byte} \times 8 \text{ bytes/message} = 256 \text{ cycles/message} = 5 \text{ seconds}$ + 5 seconds for the outbound command to get the data = 10 seconds total time.
- 8 pieces of hourly interval data for a group of 64 transponders -
 - 8 hourly interval reads/message for 64 transponders = 512 pieces of data
 - Message length for each transponder is 14 bytes (10 data + 4 overhead)
 - 6 independent orthogonal channels per phase using a CDMA type proprietary coding scheme means 6 bits of data in 4 cycles instead of 1 bit
 - $4 \text{ cycles} \times 8 \text{ bits/byte} \times 14 \text{ bytes/message} = 448 \text{ cycles/message} \times 6 \text{ channels} = 6 \text{ messages}$ in each 9 second time slot
 - 11 time slots are grouped together with 6 channels each to bring back 64 messages in $11 \times 9 = 99 \text{ seconds}$ + 5 seconds for the outbound command = 104 seconds



Achieving high throughput through concurrent phase and concurrent feeder concurrent bus communications

- 512 interval data reads from 64 transponders is retrieved by the system in 1 minute 44 seconds by using orthogonal channels, group addressing, and timeslot responses vs. 1 read in 10 seconds for a single on-request read, an improvement of 50X!
- It doesn't stop there, this is only one phase of each feeder on each substation bus!!
- TWACS 20 protocol can retrieve messages on each phase of each feeder of each substation bus concurrently
- For a substation bus with 4 feeders - 3 phases X 4 feeders = 12 X additional throughput is obtained
- 6,144 pieces of hourly interval data are retrieved from 768 transponders on each substation bus in 1 minute 44 seconds
- For a system wide deployment consisting of 300 substation busses (estimated total population 1,500,000 meters), TWACS 20 can retrieve 1.8 million interval data reads from 230,000 transponders in 1 minute 44 seconds, or over 17,000 interval data reads per second!!!



Net Result of Concurrency (maximum theoretical)

50 Hz

6 channels
X 3 phases
X 8 feeders
X 12.5 bits/second

= 1800 bits/second/bus

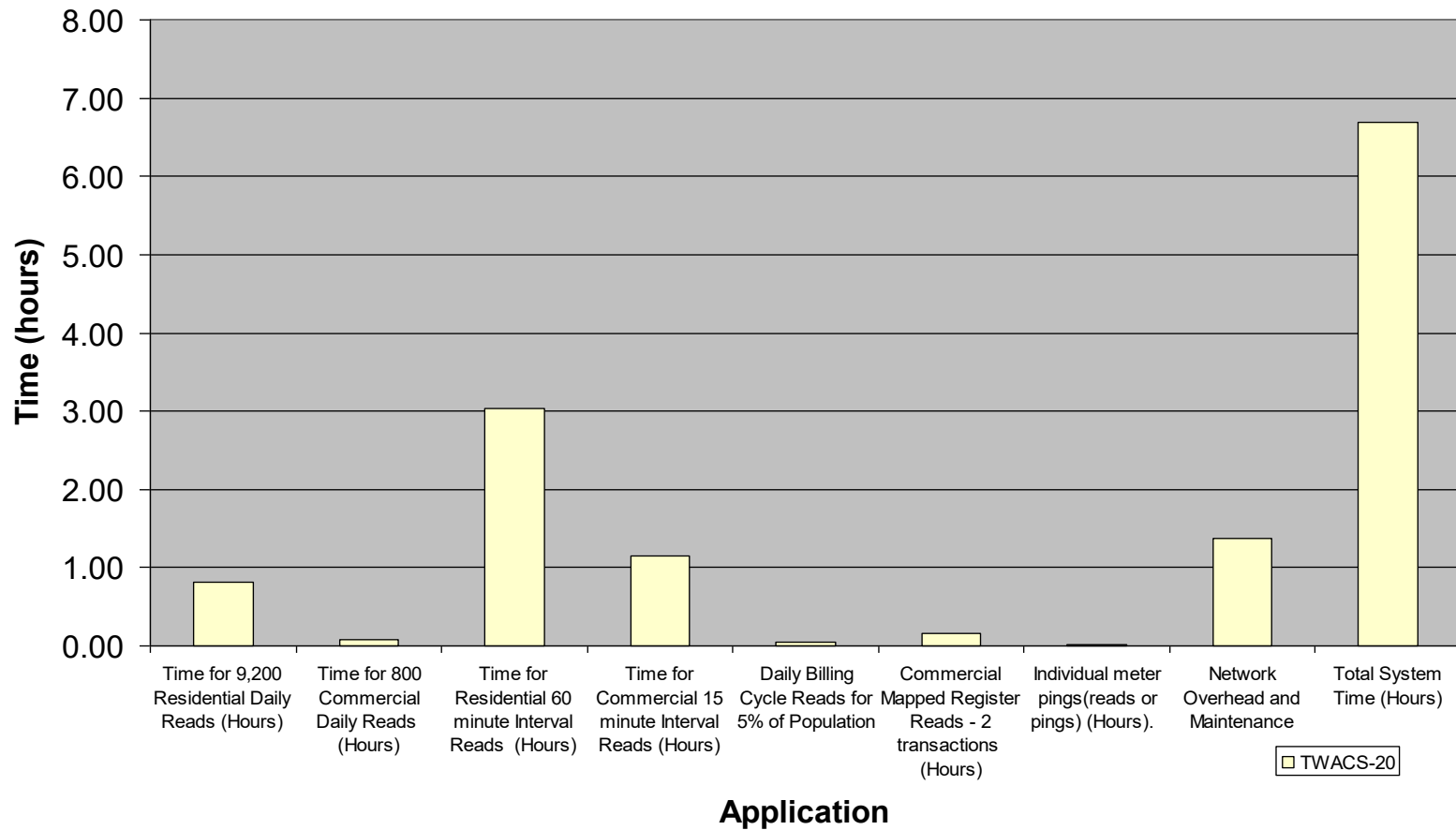
60 Hz

6 channels
X 3 phases
X 8 feeders
X 15 bits/second

= 2160 bits/second/bus



10,000 meters per bus expected times





But calculations are no substitute for the real thing:

PPL reads 1.5 million hourly interval meters
4 times per day = 6 million daily reads with a
system performance greater than 98.5%.

No other AMR/AMI Company has this proven
track record!



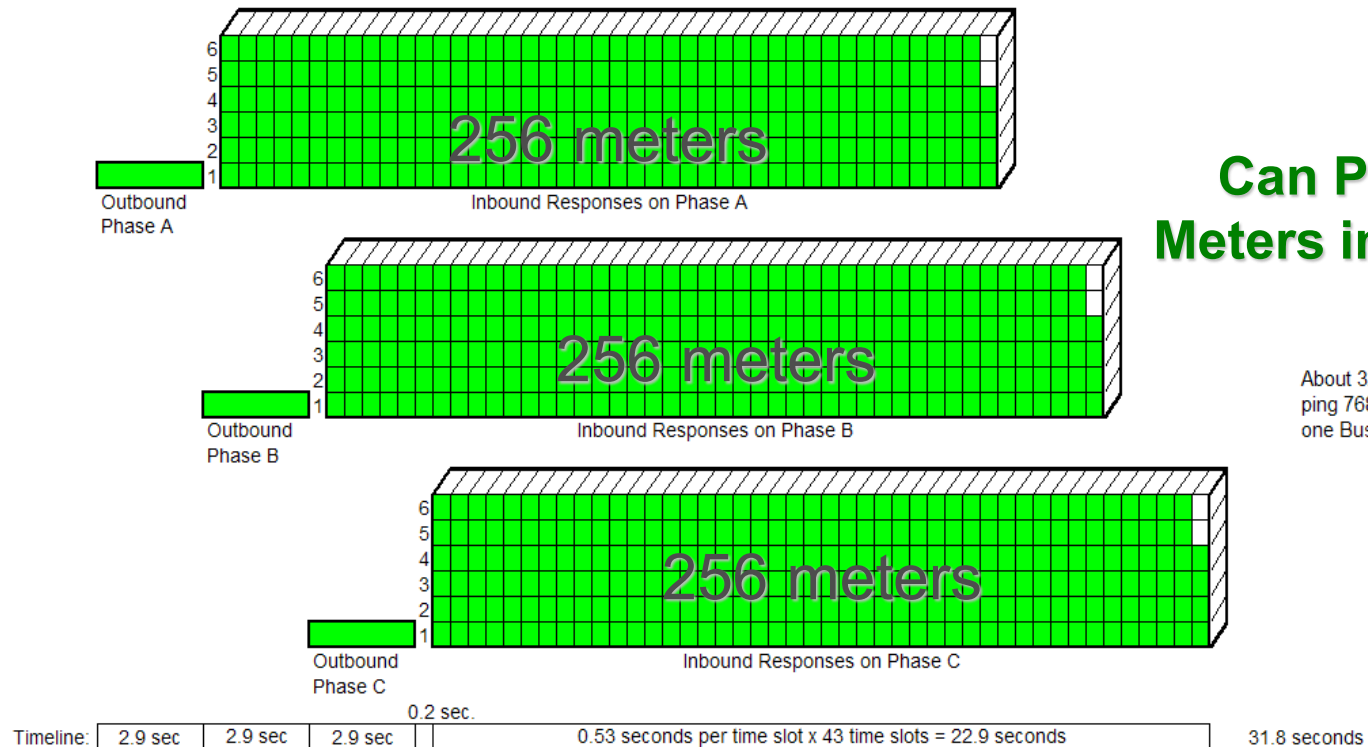
Outage Analysis and Assessment

- TWACS NG provides the capabilities with a software tool that automates outage assessment by checking the status of power at each endpoint.
 - Results can be consolidated and automatically set to a mapping or OMS system via a CIM interface for real time display.
 - Data is gathered quickly and in time to assist crews working in the field to know where power needs to be restored.
 - Also has the capability to obtain voltage information depending upon the capability of the endpoint.
- 2,500 endpoints on each substation bus can be polled for outages in just a few minutes.
- **In 1 hour TWACS can read consumption on 2,500 meters per each substation bus, collect the interval data, and ping all units for outages. For 100 busses, the same information can be retrieved for 250,000 meters in the same 1 hour!**



Larger Sized Group Pings with Concurrent Phase

Able to Analyze Entire Feeders in < 1 Minute!



**Can Ping 1,400*
Meters in a Minute**

About 32 seconds to
ping 768 meters on
one Bus

*Performance figures are for first try results utilizing a one-byte Ping response



Entire Feeder Outage Check

